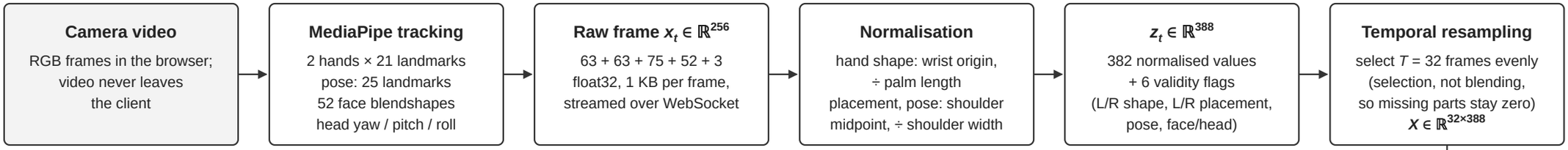
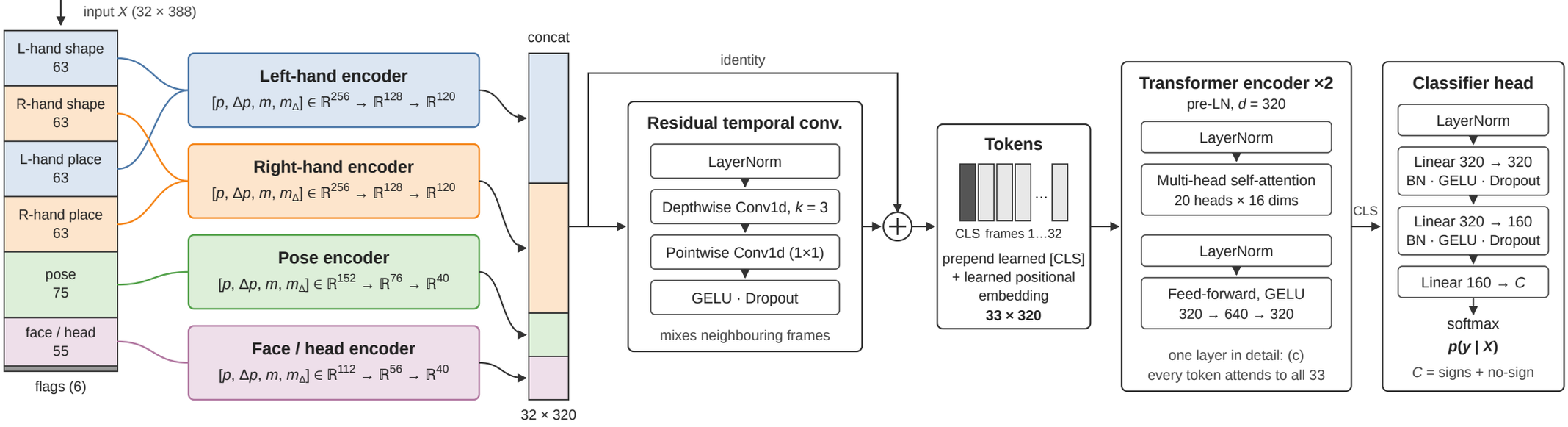
(a) Landmark extraction and normalisation (one shared implementation for training and serving)

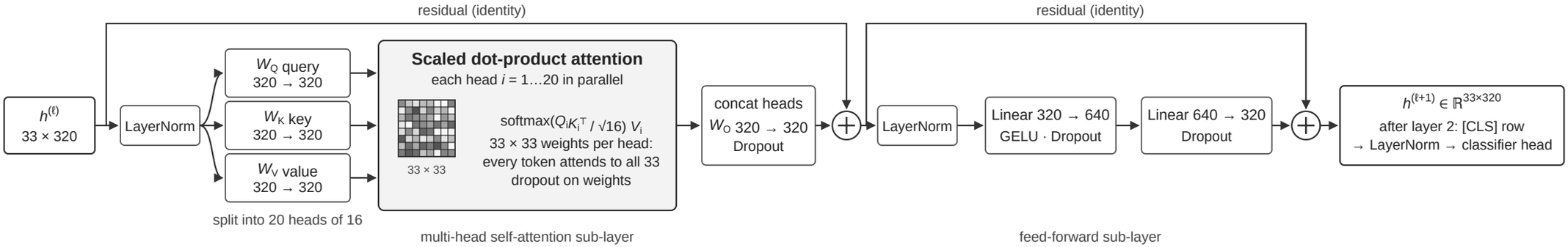


(b) SignNet classifier



each, per frame: Linear (→ input / 2) → GELU → Linear (→ region width) → GELU → LayerNorm; learned embedding e_{miss} if region absent
p: positions · Δp: frame difference, 0 unless both frames valid · m: validity flags

(c) One Transformer encoder layer (pre-LN; applied twice, separate weights)



$h' = h + \text{Dropout}(\text{MHSA}(\text{LN}(h)))$ $h^{(\ell+1)} = h' + \text{Dropout}(W_2\ \text{Dropout}(\text{GELU}(W_1\ \text{LN}(h'))))$ No attention mask: all 33 tokens are always present.

Parameters per layer: attention 410,880 (Q, K, V, O: 4 × (320 × 320 + 320)) + feed-forward 410,560 + two LayerNorms 1,280 = 822,720; both layers 1,645,440.

Trainable parameters: 2,037,668 + 161·C (2,053,929 for 100 signs + no-sign). Embedding width d = 320 = 120 + 120 + 40 + 40 (hands, pose, face/head). All dropout p = 0.1. Exported to ONNX and run on CPU.